

Tutorial 5: Three Phase Rectifiers

1. Fig. 1 shows a three pulse uncontrolled mid-point rectifier. L_p is the source inductance. The highly inductive load has been modelled as a constant current source I_o . Assume a three-phase balanced source RYB with $v_{RN} = V_m \sin(\omega t)$.

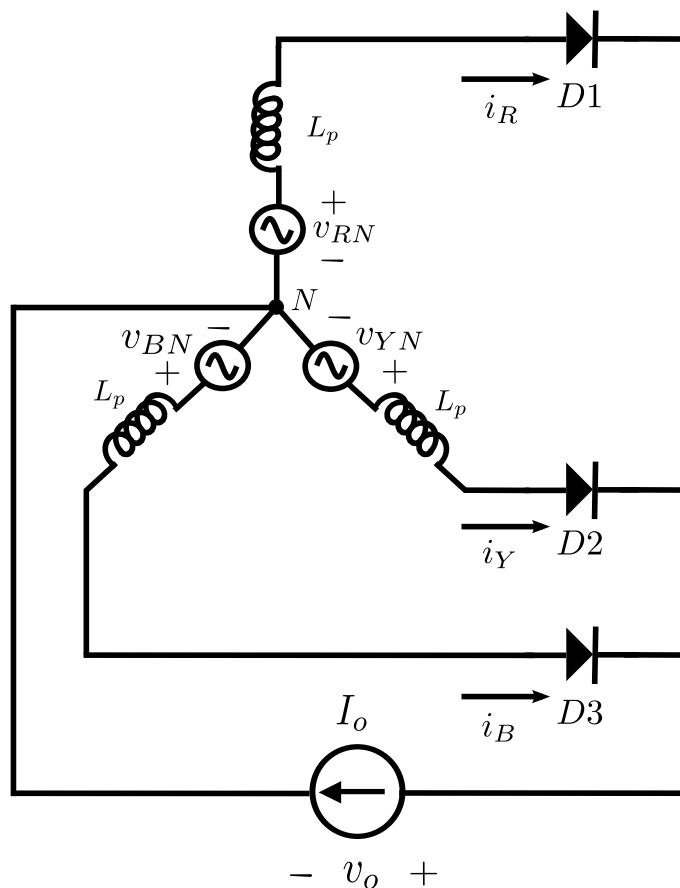


Fig. 1: Three pulse uncontrolled mid-point rectifier.

For the three pulse mid-point rectifier shown above,

- Sketch the output voltage v_o when $L_p = 0$. Indicate the devices conducting during the various sub-intervals of one fundamental cycle.
- Compute the average value of the output voltage v_o in terms of V_m when $L_p = 0$.
- Sketch the output voltage v_o when $L_p \neq 0$ assuming an overlap angle $u = 15^\circ$. Derive the output voltage and commutating diode's current during the overlap period of any one set of two devices. Indicate the devices conducting during the various sub-intervals of one fundamental cycle.
- For $L_p \neq 0$, compute the average value of the output voltage v_o in terms of V_m , I_o , ω and L_p .

2. Fig. 2 shows a three-phase semi-controlled rectifier with a free-wheeling diode. The highly inductive load has been modelled as a constant current source I_o . Assume a three-phase balanced source RYB with $v_{RN} = V_m \sin(\omega t)$. Source inductance can be neglected.

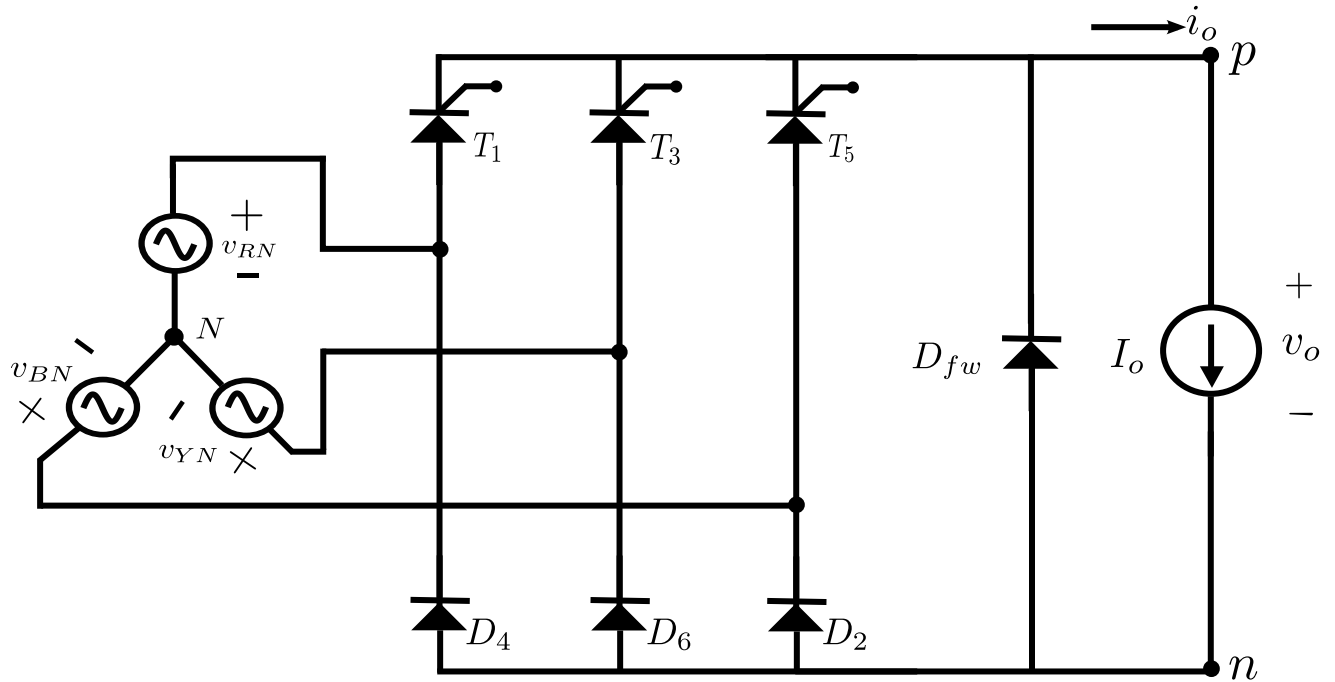


Fig. 2: Three phase semi-controlled rectifier with free-wheeling diode.

For the three-phase semi-controlled rectifier with free-wheeling diode shown above,

- Identify the valid range of firing angle α .
- Sketch the output voltage for firing angles $\alpha = 30^\circ$ and $\alpha = 90^\circ$. Indicate the conducting devices in each case. What will happen if the free-wheeling diode D_{fw} is not present?
- Derive the expression for the average value of the output voltage v_o in terms of V_m and α .